March

Day. 35

Extended ACL

Advantages of named ACL config mode.

- We can easily delete individual entries in the ACL with "no entry-number".

Eg:

```
R1(config-std-nacl)# do sh access-lists
    10 deny 1.1.1.1
    20 also permit 192.168.1.2
    30 permit 192.168.1.2
    40 permit any
R1(config-std-nacl)# no 30
Now on sh access-lists
    10 deny 1.1.1.1
    20 permit 192.168.1.2
    40 permit any
```

But if config is done with numbered ACL, if we wanna delete an entry whole ACL is deleted So, we can't delete individual entries, we only delete the entire ACL.

- We can insert new entries in between other entries by specifying the sequence number.

Eg: on earlier ACL, 10, 20, 40. we can now do

```
R1(config-std-nacl)# 30 deny 192.168.2.0 0.0.0.255
```

Resequencing ACLs

- There is a resequencing fn that helps edit ACLs.
- The command is "ip access-list resequence acl-id starting-seq-nu increment."

Eg: If seq no. are 1,2,3,4,5 we can make them 10,20,30,40,50 by below command.

R1 (config)# ip access-list resequence 1 10 10

- This works for named/numbered standard/extended ACLs.
- i.e resequencing works for all ACLs.

Extended ACLs

- Extended ACLs function mostly the same as standard ACLs.
- They can be numbered or named, just like standard ACLs
  - ↳ Numbered ACLs use the following ranges: 100-199, 2000-2699.
- They are processed from top to bottom, just like standard ACLs.
- However, they can match traffic based on more parameters, so they are more precise(& complex) then standard ACLs.
- We'll focus on matching based on these main parameters: Layer 4 protocol/port, source address, and destination address.

(numbered) R1(config)# access-list number [permit/deny] protocol src-ip dest-ip

(named) R1 (config)# ip access-list extended $name/number$.

R1 (config-ext-nacl) # [seq-num) [permit/deny] protocol src-ip dst-ip

Some protocol no.

1: ICMP 6: TCP 17: UDP 88: EIGRP 89: OSPF

We can use name also instead of number. There is also protocol named "ip" which ~~blocks~~ blocks/permit all traffic.

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Matching the src/dst IP add

- In extended ACLs, to specify a /32 addr src or dst we have to use the host option or specify the wildcard mask.
  We can't just write the add. without either of these

Eg Some examples

i) deny tcp any any
   denies all packets that encapsulate a tcp segment

ii) allow all traffic: permit ip any any

iii) prevent 10.0.0.0/16 from sending UDP traffic to 192.168.1.1/32
   # deny udp 10.0.0.0 0.0.255.255 host 192.168.1.1

iv) Prevent ~~192.16~~ 172.16.1.1/32 from pinging hosts in 192.168.0.0/24
   R1 (config-ext-nacl)# deny icmp host 172.16.1.1 192.168.0.0 0.0.0

Matching TCP/UDP port numbers

- When matching TCP/UDP, you can optionally specify the src and/or dst port numbers to match.
  command without port nos.
  R1(config-ext-nacl)# deny tcp src-ip ... dest-ip ...
  with port nos
  R1(config-ext-nacl)# deny tcp src-ip eq src-port-num dst-ip eq dst

eq 80 = equal to
gt 80 = greater than 80 (81 & greater)
lt 80 = less than 80 (79 & less)
neq 80 = NOT 80

gt
lt
neq range

gt
lt
neq range

- range 80 100 = from port 80 to port 100

- deny all packets destined for IP addr. 1.1.1.1/32, TCP port 80.
  R1(config-std-nacl) # deny tcp any host 1.1.1.1 eq 80

NOB: After the dst IP addr or dst port numbers, there are many more options we can use to match (not necessary for the CCNA).

Some examples:

- ack : match the TCP ACK flag
- fin : match the TCP FIN flag

- If we specify the protocol, src IP/port, dst IP/port etc. a packet must match all of those values to match the ACL entry. Even if it matches all except one of the parameters, the packet won't match that entry of the ACL.

Some practice qstns.

1) Allow traffic from 10.0.0.0/16 to access the server at 2.2.2.2/32 using HTTPS.
R1(config-ext-nacl) # permit tcp 10.0.0.0 0.0.255.255 2.2.2.2 0.0.0.0 eq 443.

2) Allow hosts in 172.16.1.0/24 using a TCP src port greater than 9999 to access all TCP port on server 4.4.4.4/32 except port 23
R1(config-ext-nacl) # permit tcp 172.16.1.0 0.0.0.255 gt 9999 host 4.4.4.4 neq 23.

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NOB: Extended ACLs should be applied as close to the source as possible, to limit how far the packets travel in the network before being denied. (Standard ACLs are less specific, so if they are applied to the source there's a risk of blocking more traffic than intended.)

Some useful command

- To check which ACLs are applied to an interface.

R1 ~~(config)~~ R1 # show ip int g0/0.
Outgoing access list is not set
Inbound access list is R ACCESS-LIST-NAME